AWS MLA-C02 • BLA 04 • PART 1

Deployment Infrastructure and Inference Strategies

Deployment and Orchestration of ML and AI Workflows

01

THEORY-ONLY LEARNING SERIES

Deploy • Provision • Automate



Deployment starts with how predictions are used

The consumption pattern matters more than the algorithm at this stage.

Request pattern

Real-time, near-real-time, asynchronous, or offline batch predictions.

- Who calls the model?
- How fast must it answer?
- How large is each request?

Model type

Traditional ML, foundation models, RAG systems, or agents.

- Framework and container needs
- CPU or GPU requirement
- Model size and memory

Traffic shape

Steady, spiky, intermittent, or scheduled workloads.

- Peak versus average load
- Tolerance for cold starts
- Cost per prediction



SageMaker AI offers four inference options

Match the option to latency, payload size, and traffic pattern.

Real-time endpoints

Persistent instances for low-latency, interactive predictions.

- Always-on capacity
- Supports auto scaling
- Best for steady traffic

Serverless inference

AWS manages capacity and scales with incoming requests.

- Intermittent traffic
- Pay per use
- Expect cold starts

Asynchronous inference

Queues requests and writes results to Amazon S3.

- Payloads up to 1 GB
- Processing up to 1 hour
- Can scale to zero

Batch transform

Scores a full dataset with no persistent endpoint.

- Offline scoring
- Pay only while running
- Large datasets



One endpoint can host more than one model

Sharing infrastructure reduces cost when models are small or rarely used.

Multi-model endpoints

Many models share one container and fleet, loading from S3 on demand.

- Same framework for all
- Per-customer models
- First call can be slower

Multi-container endpoints

Up to 15 different containers on one endpoint.

- Mix frameworks
- Invoke one directly
- Or chain them serially

Inference components

Assign CPU, GPU, and memory to each model on shared instances.

- Scale models independently
- Pack several FMs together
- Better GPU utilization



Foundation model hosting: managed or self-hosted

Amazon Bedrock removes infrastructure work; SageMaker AI gives deeper control.

Amazon Bedrock

- Serverless API access to many FM families
- On-demand pricing per token
- Provisioned Throughput for reserved capacity
- Custom Model Import for supported open weights

SageMaker AI hosting

- JumpStart and open-weight models
- Choose GPU instance type and count
- Control containers and serving settings
- Team owns scaling and capacity planning



Models built outside AWS can still run on AWS

Package the artifacts, choose the serving path, then validate.

1 — Package

Store model artifacts as model.tar.gz in Amazon S3

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2 — Containerize

Use a framework container or an ECR image

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3 — Import

Create a SageMaker model or Bedrock import job

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4 — Deploy

Create the endpoint configuration and endpoint

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5 — Validate

Test outputs, latency, and cost

Trained on-premises or in another cloud? The artifacts and the container are what matter, not where training happened.



Agents and RAG need deployment decisions too

Generative AI systems combine models with tools, data, and memory.

Agents

Amazon Bedrock Agents and Amazon Bedrock AgentCore run task-focused agents.

- Action groups call APIs
- Instructions define the role
- Guardrails limit output

Agent communication

Standard protocols connect agents to tools and to each other.

- MCP exposes tools and data
- A2A links agents together
- Gateways manage tool access

RAG configuration

Retrieval settings decide which context reaches the model.

- Chunk size and overlap
- Semantic or hybrid search
- Reranking and filters



A deployment decision checklist

Choose the lightest infrastructure that meets the service level.

1. Define latency, payload size, and traffic pattern.
2. Pick real-time, serverless, asynchronous, or batch inference.
3. Decide whether models should share an endpoint.
4. Choose Amazon Bedrock or SageMaker AI for foundation models.
5. Plan agent tools, protocols, and RAG retrieval settings.
6. Validate quality, latency, and cost before go-live.

